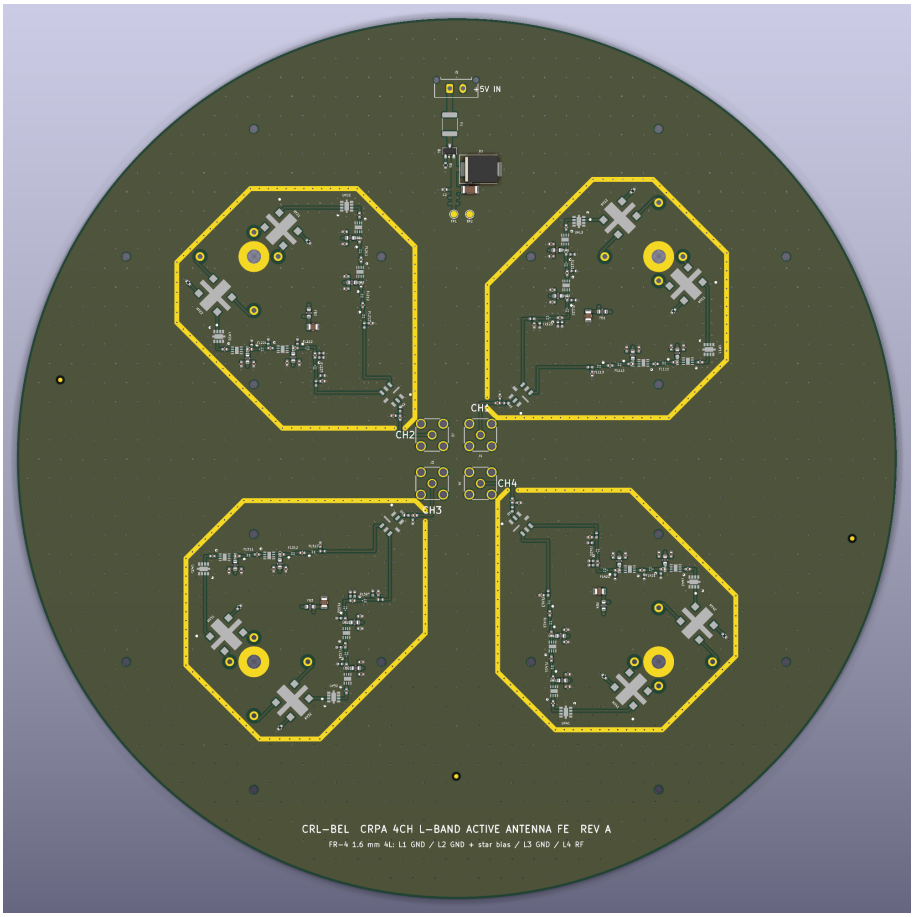
CRPA 4-Channel L-Band Active Antenna Front End - PCB Design Report

Project 17, Rev D, 2026-11-02. Board: KiCad/CRPA17_4CH_LBAND.kicad_pcb (KiCad 10). Generated by Scripts/make_docs.py; every number below is regenerated from the design data.

1. Summary

Four identical RHCP channels on the customer's 200 mm round CRPA board. Antenna module positions (92 mm square pitch, sequential 90 deg rotation), feed-pin positions and the board outline are copied from the customer Gerber (Input/Gerber_4CH_GNSS.zip). Per channel and band: patch 0/90 feeds -> HC125A hybrid -> CLM-83-2W+ limiter -> TQP3M9037 -> SAW -> TQP3M9037 -> SAW -> 100 pF -> SAW, then DXP.02.A combines L1 and L2/L5 onto one SMA output per channel. The chain, values and matching are those of project 16; the limiter (Rev A) and the third SAW per band before the diplexer (Rev B) are new. All LNAs run from one +5 V input (J5). Only the L-band is built (no NavIC S-band), as requested.

Item	Result
Board	Ø200 mm, 4-layer FR-4 1.6 mm, ENIG, all SMD on the bottom (L4), antennas on top
Channels	4 x (L1 + L2/L5) -> 4 SMA outputs (J1..J4) at the board centre
Bias	single +5 V point J5 (Molex Micro-Fit), PTC + reverse-polarity P-FET + TVS; star distribution on L2: one trunk + ferrite + 10 uF per channel
Shielding	one soldered shield can per channel (octagonal frame land, GND via fence, RF exit notch), as on the customer product
Antenna	Taoglas GPDF6010.A: footprint, drills, clearances and grounded centre screw per its data sheet
Parts	233 placed footprints, 229 fitted (29 BOM lines)
DRC (kicad-cli, zones refilled)	0 violations, 0 unconnected
Channel matching	4 channels are one layout rotated 4x: identical track lengths per net; hybrid feed pairs equal (L1 2.718/2.718 mm, L2 7.082/7.082 mm)

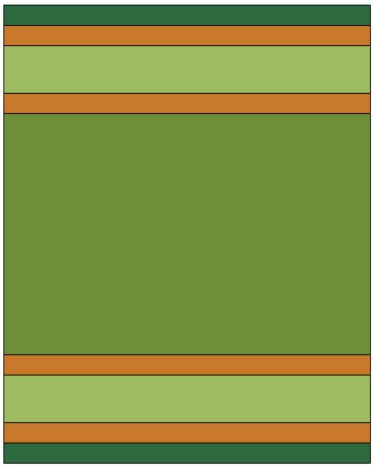


2. Design decisions

Topic	Decision	Why
Substrate	1.6 mm (not 0.8 mm), 4 layers, FR-4 7628 build	A Ø200 mm board carrying 4 antenna modules needs the stiffness of 1.6 mm. With 4 layers the RF reference (L3) is only 0.21 mm under the traces, so 50 ohm = 0.38 mm, close to the 0402 / DFN / SAW pad widths. On a 2-layer 0.8 mm board 50 ohm would be ~1.5 mm and on 1.6 mm ~3 mm, much wider than every pad (large steps), and the bias would have to cut the ground.
Layer use	L1 GND (antenna ground plane), L2 GND + per-channel +5V star traces, L3 GND, L4 RF + parts	Solid ground under the patches as on the original board; bias never crosses an RF reference; every SMD part on one side (one reflow pass).
Material	FR-4 (Tg170) instead of RO4350B	Pre-LNA lines are short (see budget): FR-4 adds 0.082 dB/cm, i.e. 0.13-0.18 dB NF for the routed pre-LNA lines. Option: L3-L4 in RO4350B 0.254 mm (hybrid build) saves about half of that.
Limiter position	after the hybrid, before LNA1 (one per band per channel, 8 total)	Protects every LNA input; the hybrid (30 W) does not need protection. One limiter per band instead of one per feed halves the count.
Limiter part (Rev D)	CLM-83-2W+ (DL1721, 3x3 x 1.14 mm hermetic LTCC) replaces RLM-63-2W+	Customer choice. Same +11.5 dBm clamp and 2 W rating; IN (2) and OUT (7) face each other, so it sits in-line with no 5 mm jog; 0.13-0.16 dB IL in the GNSS bands. GND per PL-377: paddle vias + tab vias.
Input TVS (Rev D)	1.5SMC6.8A (1500 W, 5.8 V standoff, SMC) replaces SMAJ5.0A	Customer choice: 1500 W vs 400 W surge rating. The 5.8 V standoff is above the 5.25 V maximum input (SMAJ5.0A, 5.0 V, sat below it). It absorbs surges / ESD; a sustained over-voltage is left to F1 (PTC).
3rd SAW (Rev B)	one more SAW per band right before the DXP (B7504 / B8889), 100 pF between SAW 2 and SAW 3	More out-of-band rejection at the output (customer request). Each B8889 port keeps its own 6.8 nH shunt; the 100 pF gives each shunt its own pad. Costs 0.9-1.5 dB gain, no NF change.
Single bias point	dedicated DC connector J5, no DC on any RF output	A bias tee on only one channel would make that channel different (CRPA needs matched channels); 1.1-1.4 A is also too much for a small coax bias inductor. C{n}43 blocks receiver DC.
Bias distribution (Rev C)	star: 4 separate 0.8 mm trunks on L2 from the star point after Q1; per channel a ferrite (FB{n}, BLM18KG601) + 10 uF, then 0.4 mm branches to the 4 LNA chokes	Channels share nothing but the star point, so LNA leakage / noise in one channel is isolated from the others by two ferrites; L2 is otherwise solid GND (no +5V plane cavity resonances, third RF ground).
Shield cans (Rev C)	octagonal can per channel on a 1.0 mm exposed GND frame land with vias every 2 mm; 2.6 mm notch where the RF output leaves	~40 dB gain per channel and 4 channels side by side: the cans stop channel-to-channel and external coupling (the customer product uses cans too). Outline: Fab/CRPA17_ShieldCan_outline.csv, Images/shield_can_outline.png.
Antenna module (Rev C)	Taoglas GPDF6010.A (from the product photo): D3.2 centre screw hole GROUNDED, 4x D1.8 locating holes at 29 mm, feed pins D0.9 in 1.1 mm holes, D3.5 copper-free around each feed	Data sheet SPE-21-8-111-G sec. 3.1, 4.3, 4.5: the screw is the antenna ground; pins are 2.5 mm long, so they pass the 1.6 mm board.
Output connectors	4 x SMA vertical (Amphenol 132134) at the board centre	Short cable exit under the array, in the patch-free cross between modules, all channels equal.
Land patterns	HC125A, TQP3M9037, B7504, B8889, DXP.02.A, antenna: copied from the customer board; CLM-83-2W+ (DL1721): from its data sheet (PL-377)	Customer patterns are proven for their assembly; mirrored back to a top view by Scripts/build_footprints.py.
Hybrid pin mapping	kept from the customer board (L1 0->4, 90->3; L2 0->3, 90->4)	Same sense of polarisation as the original board (RHCP at pin 1).

3. Stack-up and 50 ohm line

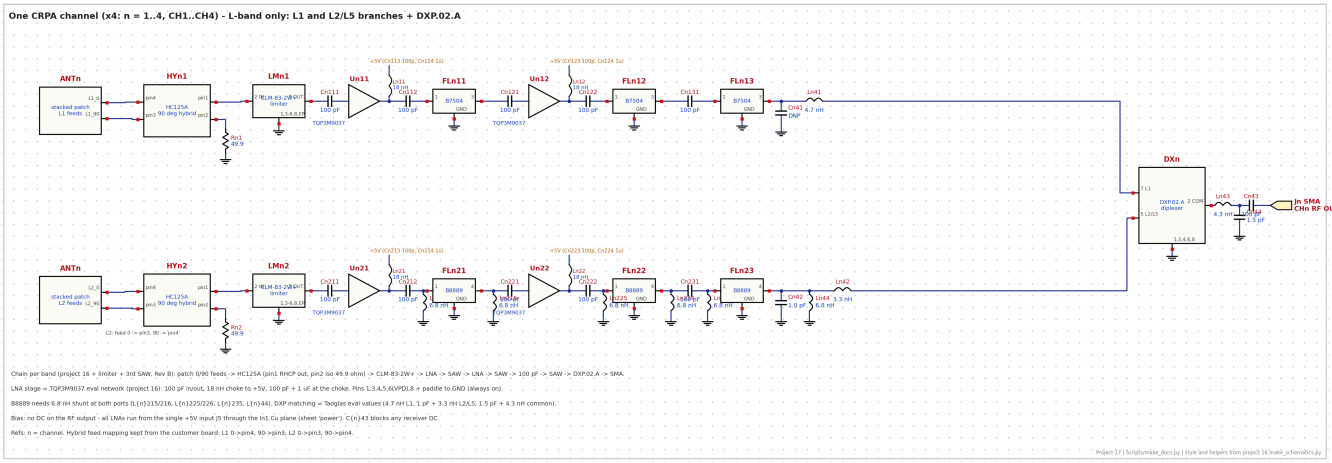
4-layer FR-4, 1.6 mm finished (JLC04161H-7628 class). 50 ohm on L4: 0.38 mm over L3, 0.30 mm gap to pour

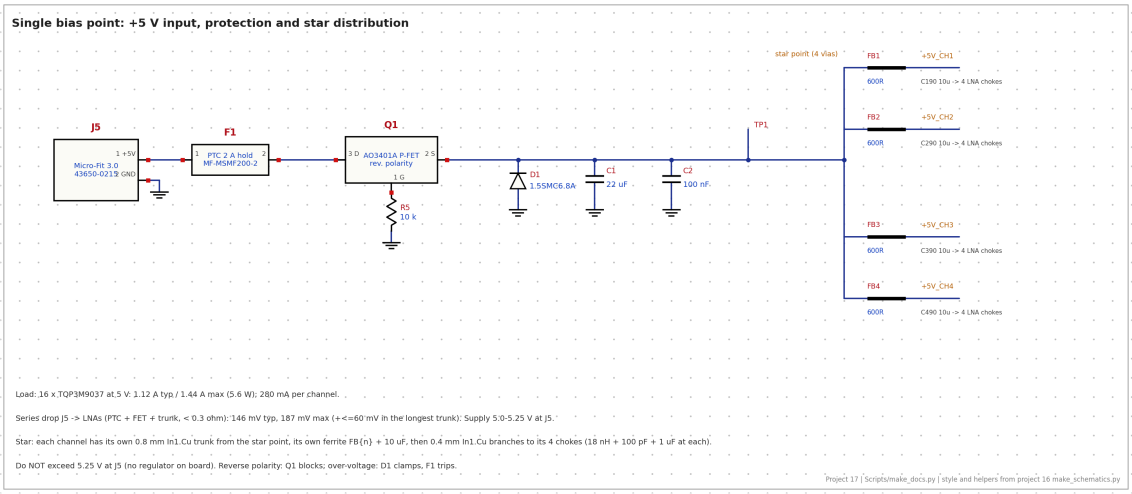


F.Mask 10 um er 3.8 green LPI
F.Cu 35 um L1: solid GND = antenna ground plane
PP1 0.210 mm er 4.4 7628 RC49%
In1.Cu 15 um L2: GND + per-channel +5V star traces
Core 1.065 mm er 4.6 FR-4 core
In2.Cu 15 um L3: solid GND, RF reference for L4
PP2 0.210 mm er 4.4 7628 RC49%
B.Cu 35 um L4: RF lines + all SMD parts
B.Mask 10 um er 3.8 green LPI

Fab note: impedance control on L4 (50 ohm +/-10 %, 0.38 mm over L3, coupon). Hammerstad-Jensen gives 49.5 ohm for 0.38 mm on 0.2104 mm (er 4.4); the copper pour keeps 0.30 mm from RF copper. Fine-pitch pads are entered with a 0.2 mm x 0.6 mm neck. GND via fences (pitch 2.5 mm, 1.05 mm from the line) along every 50 ohm segment plus a 5 mm stitching grid tie L1/L3/L4.

4. Schematic (one channel, x4) and bias





5. RF budget (project 16 chain + this board's limiter and lines)

Stage data from project 16 (Extracted_Data/results_cascade_budget.csv, verified there against heavi, circuitRF and Qucs). Added in front of LNA1: the routed line lengths of this board (from the PCB) at 0.082 dB/cm and the CLM-83-2W+ insertion loss (data sheet table, interpolated).

Case	Pre-LNA line mm	Limiter IL dB	Gain p16 dB	Gain this board dB	NF p16 dB	NF this board dB	dNF dB
L1 C/A (1575.42 MHz, 2.046 MHz)	21.71	0.165	38.45	37.21	0.742	1.085	0.343
L5 (1176.45 MHz, 20.46 MHz)	15.88	0.138	42.06	40.88	0.591	0.859	0.268
L2C (1227.60 MHz, 2.046 MHz)	15.88	0.142	40.09	38.36	0.622	0.894	0.272

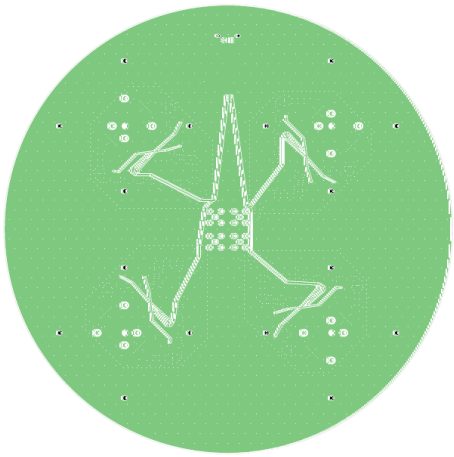
The limiter clamps at about +11.5 dBm (data sheet) - this protects the LNA against damage (strong nearby transmitters, ESD); it does not stop LNA compression, which starts near -1 dBm at the LNA input. Filtering before LNA1 (DME/Iridium) stays an open point from project 16.

6. Power

Quantity	Value
LNAs	16 x TQP3M9037 at +5 V
Current	1.12 A typ, 1.44 A max (280 mA per channel)
Dissipation	5.6 W spread over 16 parts
Drop J5 -> LNAs	146 mV typ / 187 mV max
Input range	5.0 - 5.25 V at J5 (no regulator; TQP3M9037 max 5.25 V recommended)

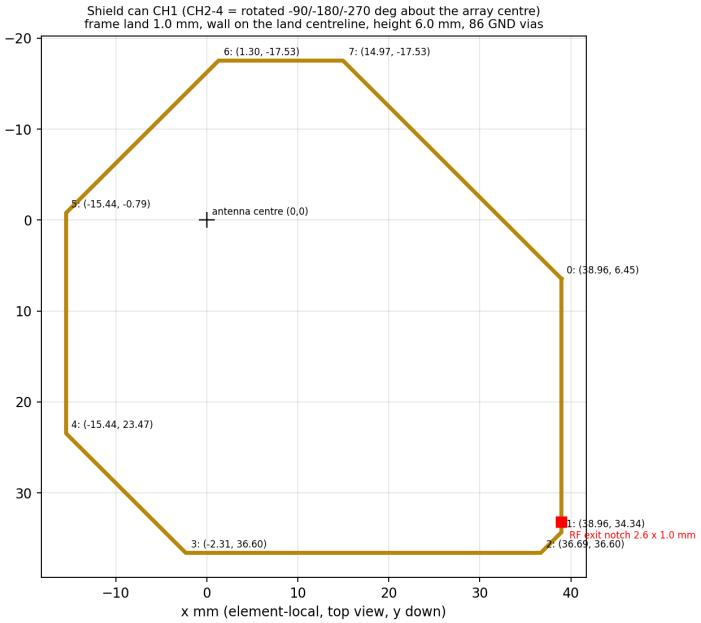
6a. Star bias distribution (In1.Cu)

J5 -> F1 -> Q1 -> star point (4 vias). From there each channel has its own 0.8 mm trunk on In1.Cu (CH1 97.5 mm, CH2 99.9 mm, CH3 118.0 mm, CH4 117.4 mm), its own ferrite FB{n} + 10 uF C{n}90, and 0.4 mm In1.Cu branches to its 4 LNA bias vias (18 nH choke + 100 pF + 1 uF at each). The rest of In1.Cu is GND. Routes were found by a grid router that keeps every through-hole clear; CH1's branches are replayed on CH2-4 so all channels stay identical. From the board: 65 bias pads (16 chokes, 4 ferrites), 28 bias vias; DRC 0 unconnected.



6b. Shield cans

One can per channel, height 6.0 mm (CLM-83-2W+ limiter 1.14 mm high; antenna nut about 2.4 mm). The frame land is 1.0 mm exposed copper (GND) with 86 GND vias per channel; the can wall sits on the land centreline. The RF output leaves through a 2.6 mm x 1.0 mm notch in the wall (red square). Order as a two-piece frame + lid (rework access) or a one-piece drawn can, tin-plated steel 0.2 mm.



7. Verification

Check	Tool	Result
DRC + zone refill	kicad-cli 10.0.3 pcb drc	0 violations, 0 unconnected
Routing completeness	kicad-happy analyze_pcb.py	complete, 0 unrouted nets
Gerber set	kicad-happy analyze_gerbers.py	4 Cu layers, complete, aligned. GR-005 'outline not closed' = false positive (the outline is a full circle drawn as 2 arcs, as on the customer Gerber). GR-004 paste = expected (no paste on THT/vias).
Channel identity	net lengths from the PCB	all 4 channels equal per net
Footprint pin maps	customer Gerber X2 pad attributes	HC125A, TQP3M9037, SAWs, DXP: same pin numbers / nets as the customer board; CLM-83-2W+ from the data sheet drawing PL-377 (pin 2 IN, 7 OUT in-line, paddle GND)

8. Assumptions and open points

#	Item	Action
R1	RESOLVED (Rev C): Taoglas GPDF6010.A data sheet - pins D0.9 / 2.5 mm long (pass 1.6 mm), holes 1.1 mm, centre screw 3.2 mm grounded, locating holes 1.8 mm, module D60.6 mm	none
A3	Future NavIC S-band patches (product photo: 4 square patches on the axes) are not on this board; the L2 star trunks and the SMA cluster use that area	Decide S-band before Rev D
A6	Antenna nut (bottom) is torqued after reflow; HC125A sits ~6 mm from the screw	Use a thin-wall socket, or fit the antenna before the shield cans
A4	FR-4 er 4.4 / tan d 0.02 for the 7628 prepreg	Fab impedance report; adjust RF_W in crpa_design.py
A5	Land patterns of HC125A, DXP.02.A, B7504, B8889, TQP3M9037 copied from the customer board	Cross-check against vendor drawings (not in the repository)
O1	No pre-LNA band filter (project 16 open point: DME 1150 MHz, Iridium 1621 MHz)	Design decision
O2	Gain ~38-42 dB per channel is high for most CRPA receivers	Add a pad at the output if needed
O3	Vias in the HC125A / TQP3M9037 GND pads	Order resin-plugged + capped vias (IPC-4761 VII) or tent them from the top

9. Bill of materials

#	Qty	Value	MPN	Mfr	Refs	Fit
1	8	HC125A	HC125A	Taoglas	HY11 HY12 HY21 HY22 HY31 HY32 HY41 HY42	SMT bottom
2	8	CLM-83-2W+	CLM-83-2W+	Mini-Circuits	LM11 LM12 LM21 LM22 LM31 LM32 LM41 LM42	SMT bottom
3	16	TQP3M9037	TQP3M9037	Qorvo	U111 U112 U121 U122 U211 U212 U221 U222 U311 U312 U321 U322 U411 U412 U421 U422	SMT bottom
4	12	B7504	B39162B7504L210	Qualcomm RF360	FL111 FL112 FL113 FL211 FL212 FL213 FL311 FL312 FL313 FL411 FL412 FL413	SMT bottom
5	12	B8889	B39122B8889P810	Qualcomm RF360	FL121 FL122 FL123 FL221 FL222 FL223 FL321 FL322 FL323 FL421 FL422 FL423	SMT bottom
6	4	DXP.02.A	DXP.02.A	Taoglas	DX1 DX2 DX3 DX4	SMT bottom
7	16	18nH	LQW18AN18NG00D	Murata	L111 L112 L121 L122 L211 L212 L221 L222 L311 L312 L321 L322 L411 L412 L421 L422	SMT bottom
8	24	6.8nH	LQW15AN6N8G00D	Murata	L144 L244 L344 L444 L1215 L1216 L1225 L1226 L1235 L2215 L2216 L2225 L2226 L2235 L3215 L3216 L3225 L3226 L3235 L4215 L4216 L4225 L4226 L4235	SMT bottom
9	4	4.7nH	LQW15AN4N7B00D	Murata	L141 L241 L341 L441	SMT bottom
10	4	3.3nH	LQW15AN3N3B00D	Murata	L142 L242 L342 L442	SMT bottom
11	4	4.3nH	LQW15AN4N3B00D	Murata	L143 L243 L343 L443	SMT bottom
12	60	100pF	GRM1555C1H101JA01D	Murata	C143 C243 C343 C443 C1111 C1112 C1113 C1121 C1122 C1123 C1131 C1211 C1212 C1213 C1221 C1222 C1223 C1231 C2111 C2112 C2113 C2121 C2122 C2123 C2131 C2211 C2212 C2213 C2221 C2222 C2223 C2231 C3111 C3112 C3113 C3121 C3122 C3123 C3131 C3211 C3212 C3213 C3221 C3222 C3223 C3231 C4111 C4112 C4113 C4121 C4122 C4123 C4131 C4211 C4212 C4213 C4221 C4222 C4223 C4231	SMT bottom
13	16	1uF	GRM155R61A105KE15D	Murata	C1114 C1124 C1214 C1224 C2114 C2124 C2214 C2224 C3114 C3124 C3214 C3224 C4114 C4124 C4214 C4224	SMT bottom
14	4	1.0pF	GJM1555C1H1R0BB01D	Murata	C142 C242 C342 C442	SMT bottom
15	4	1.5pF	GJM1555C1H1R5BB01D	Murata	C144 C244 C344 C444	SMT bottom
16	4	10uF	GRM21BR61A106KE19L	Murata	C190 C290 C390 C490	SMT bottom
17	1	22uF	GRM31CR61A226KE15L	Murata	C1	SMT bottom
18	1	100nF	GRM155R71C104KA88D	Murata	C2	SMT bottom
19	8	49.9R	ERJ-2RKF49R9X	Panasonic	R11 R12 R21 R22 R31 R32 R41 R42	SMT bottom
20	1	10k	ERJ-2RKF1002X	Panasonic	R5	SMT bottom
21	1	2A hold	MF-MSMF200-2	Bourns	F1	SMT bottom
22	1	AO3401A	AO3401A	Alpha & Omega	Q1	SMT bottom
23	1	1.5SMC6.8A	1.5SMC6.8A	Littelfuse	D1	SMT bottom
24	4	SMA	132134	Amphenol RF	J1 J2 J3 J4	Hand solder (THT)

#	Qty	Value	MPN	Mfr	Refs	Fit
25	1	+5V IN	43650-0215	Molex	J5	Hand solder (THT)
26	2	TP	-	-	TP1 TP2	SMT bottom
27	4	DNP	-	-	C141 C241 C341 C441	DNP
28	4	L1/L2 stacked patch	existing module	customer	ANT1 ANT2 ANT3 ANT4	Customer supplied, hand fit (top)
29	4	600R@100MHz	BLM18KG601SN1D	Murata	FB1 FB2 FB3 FB4	SMT bottom

10. Files

Path	Content
Fab/CRPA17_4CH_LBAND_Gerbers.zip	Gerber X2 + Excellon (PTH/NPTH) + drill map + job file
Fab/CRPA17_4CH_LBAND_BOM.csv / .xlsx	BOM
Fab/CRPA17_4CH_LBAND_CPL_bottom.csv	pick-and-place (bottom side, mm, deg)
Fab/CRPA17_4CH_LBAND.d356	IPC-D-356 netlist (bare-board test)
KiCad/	board, project, DRC rules, footprint library CRPA17.pretty
Docs/	this report, layer PDF, netlist, RF budget CSVs
Scripts/	build_footprints.py, build_pcb.py, run_drc.sh, export_fab.sh, make_docs.py, build_all.sh